NAME

mbphotogrammetry − Generate bathymetry soundings from navigated stereo pairs of downlooking seafloor photographs.

VERSION

Version 5.0

SYNOPSIS

mbphotogrammetry [**−−verbose** **−−help** **−−threads=***nthreads* **−−show−images** **−−input=***imagelist* **−−fov−fudgefactor=***factor* **−−navigation−file=***file* **−−survey−line−file=***file* **−−tide−file=***file* **−−image−quality−file=***file* **−−image−quality−threshold=***value* **−−image−quality−filter−length=***value* **−−output=***fileroot* **−−output−number−pairs=***value* **−−camera−calibration−file=***file* | **−−calibration−file=***file* **−−platform−file=***platform.plf* **−−camera−sensor=***id* **−−nav−sensor=***id* **−−sensordepth−sensor=***id* **−−heading−sensor=***id* **−−altitude−sensor=***id* **−−attitude−sensor=***id* **−−altitude−min=***value* **−−altitude−max=***value* **−−trim=***value* **−−bin−size=***value* **−−bin−filter=***value* **−−downsample=***value* **−−algorithm=***algorithm* **−−algorithm−pre−filter−cap=***value* **−−algorithm−sad−window−size=***value* **−−algorithm−smoothing−penalty−1=***value* **−−algorithm−smoothing−penalty−2=***value* **−−algorithm−min−disparity=***value* **−−algorithm−number−disparities=***value* **−−algorithm−uniqueness−ratio=***value* **−−algorithm−speckle−window−size=***value* **−−algorithm−speckle−range=***value* **−−algorithm−disp−12−max−diff=***value* **−−algorithm−texture−threshold=***value*]

DESCRIPTION

Mbphotogrammetry generates bathymetry soundings from navigated stereo pairs of downlooking photographs taken from a submerged survey platform. For each stereo pair listed in an input MB-System imagelist file, the program computes a dense disparity map between the (optionally undistorted and rectified) left and right images using one of OpenCV’s block matching stereo correspondence algorithms, then converts the resulting disparities into 3D points using the stereo camera’s calibrated geometry. These 3D points are combined with the platform’s navigation, heading, and attitude at the time each stereo pair was taken (derived directly from the imagelist and, optionally, an MB-System platform offset model and separate navigation and tide files) to compute georeferenced soundings, which are binned and written out as MB-System multibeam-format soundings (format 251, *mb251*, associated with the *mbsys_stereopair* data structure).

Three OpenCV stereo correspondence algorithms are supported: block matching (**bm**), semi-global block matching (**sgbm**, the default), and its more exhaustive variant (**hh**). Numerous options tune the chosen algorithm’s matching window, disparity range, smoothing penalties, and validity checks; these correspond directly to the parameters of OpenCV’s **StereoBM** and **StereoSGBM** classes. Soundings whose stereo-derived altitude falls outside the range set by **−−altitude−min** and **−−altitude−max** are rejected. Output soundings can optionally be binned spatially, with each bin’s disparity resolved either by taking the mean or the median of the pixels falling in that bin.

Input images can be filtered by an externally computed per-image quality metric (**−−image−quality−file** together with **−−image−quality−threshold**), and processing can be parallelized across multiple stereo pairs at once with **−−threads**. Output soundings are normally written to a single file named from **−−output** with the suffix *.mb251*, but can instead be split into a new output file every **−−output−number−pairs** stereo pairs, or split at the waypoints of a survey line file supplied with **−−survey−line−file**.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−help

This "help" flag causes the program to print out a description of its operation and then exit immediately.

−−verbose

The **−−verbose** option causes **mbphotogrammetry** to print out status and debug messages. Using it more than once increases the verbosity level.

−−threads=*nthreads*

Sets the number of processing threads used to work through the stereo pairs in the image list. The requested number of threads is limited to the number of hardware threads available and to the MB-System maximum thread count. The default is 1 thread.

−−show−images

Enables display of the images being processed (via OpenCV's image display windows) as the program runs.

−−input=*imagelist*

Sets the filename of the input MB-System imagelist file listing the stereo image pairs, and their navigation, attitude, and camera settings, to be processed. The default input filename is *imagelist.mb−1*.

−−fov−fudgefactor=*factor*

Sets a multiplicative fudge factor applied to the camera field of view angles. The default value is 1.0 (no adjustment).

−−navigation−file=*file*

Sets the filename of an ASCII navigation file used in place of, or to supplement, navigation carried in the imagelist file.

−−survey−line−file=*file*

Sets the filename of a survey line/route file whose waypoint times are used to split the output soundings into a separate *.mb251* output file at each waypoint crossed.

−−tide−file=*file*

Sets the filename of an ASCII tide file (time and tide height pairs) used to apply a tidal correction to the computed soundings.

−−image−quality−file=*file*

Sets the filename of an ASCII file of time-stamped image quality values used to filter out low quality stereo pairs.

−−image−quality−threshold=*value*

Sets the minimum image quality value (in the range 0 to 1) required for a stereo pair to be processed.

−−image−quality−filter−length=*value*

Sets a time filter length, in seconds, applied when interpolating image quality values from the image quality file.

−−output=*fileroot*

Sets the root name used to construct the output sounding file name(s). Unless **−−survey−line−file** or **−−output−number−pairs** is used to split the output into multiple files, all soundings are written to a single file named *fileroot***.mb251**.

−−output−number−pairs=*value*

Splits the output soundings into a new file, named *fileroot_nnn***.mb251**, every time *value* stereo pairs have been written to the current output file. This option is ignored if **−−survey−line−file** is also used.

−−camera−calibration−file=*file*

−−calibration−file=*file*
These two equivalent options set the filename of an OpenCV YAML file containing the intrinsic and extrinsic stereo camera calibration parameters (camera matrices, distortion coefficients, and the rotation/translation between the two cameras) used to undistort and rectify stereo pairs that are not already rectified, and to convert stereo disparities into 3D coordinates.

−−platform−file=*platform.plf*
Sets the filename of an MB-System platform definition file describing the geometric offsets between the vehicle's navigation, attitude, and camera sensors.

−−camera−sensor=*camera_sensor_id*
Sets the platform sensor id number of the stereo camera used to look up the camera's position and orientation offsets within the platform model.

−−nav−sensor=*nav_sensor_id*
Overrides the platform model's navigation (position) source sensor id number.

−−sensordepth−sensor=*sensordepth_sensor_id*
Overrides the platform model's pressure/depth source sensor id number.

−−heading−sensor=*heading_sensor_id*
Overrides the platform model's heading source sensor id number.

−−altitude−sensor=*altitude_sensor_id*
Sets the platform sensor id number used as the altitude source.

−−attitude−sensor=*attitude_sensor_id*
Overrides the platform model's roll/pitch and heave source sensor id number.

−−altitude−min=*value*
Sets the minimum acceptable stereo-derived altitude, in meters, for a sounding to be retained. The default is 1.0 meter.

−−altitude−max=*value*
Sets the maximum acceptable stereo-derived altitude, in meters, for a sounding to be retained. The default is 5.0 meters.

−−trim=*value*
Sets a pixel trim margin excluded around the edges of each stereo image before computing disparity. The default is 0.0 (no trim).

−−bin−size=*value*
Sets the size, in pixels, of the spatial bins used to combine disparity values before generating output soundings. The default is 1 (no binning).

−−bin−filter=*value*
Selects how disparity values are combined within each spatial bin: **mean** (or **0**) averages the values in the bin, while **median** (or **1**) takes the median value. The default is **mean**.

−−downsample=*value*
Sets a downsampling factor applied to the stereo images before disparity matching. The default is 1 (no downsampling).

−−algorithm=*algorithm*
Selects the OpenCV stereo correspondence algorithm used to compute disparities. Valid values are **bm** (block matching), **sgbm** (semi-global block matching, the default), and **hh** (a more exhaustive variant of semi-global block matching).

−−algorithm−pre−filter−cap=*value*
Sets the truncation value for prefiltered image pixels used by the stereo correspondence algorithm. The default is 4.

−−algorithm−sad−window−size=*value*
Sets the matched block (sum of absolute differences window) size used by the stereo correspondence algorithm. The default is 5.

−−algorithm−smoothing−penalty−1=*value*
Sets the first smoothness penalty parameter (P1) controlling disparity change of one pixel between neighboring pixels, used by the semi-global block matching algorithms. The default is 600.

−−algorithm−smoothing−penalty−2=*value*
Sets the second smoothness penalty parameter (P2) controlling larger disparity changes between neighboring pixels, used by the semi-global block matching algorithms. The default is 2400.

−−algorithm−min−disparity=*value*
Sets the minimum possible disparity value used by the stereo correspondence algorithm. The default is -64.

−−algorithm−number−disparities=*value*
Sets the number of disparity values searched by the stereo correspondence algorithm (the disparity search range). The default is 192.

−−algorithm−uniqueness−ratio=*value*
Sets the margin, in percent, by which the best matching disparity cost must beat the next best value for a match to be considered valid. The default is 1.

−−algorithm−speckle−window−size=*value*
Sets the maximum size of smooth disparity regions considered noise speckles and invalidated by the stereo correspondence algorithm's postfiltering. The default is 150.

−−algorithm−speckle−range=*value*
Sets the maximum disparity variation within a connected component considered during speckle postfiltering. The default is 2.

−−algorithm−disp−12−max−diff=*value*
Sets the maximum allowed difference, in integer pixel units, between the left-to-right and right-to-left disparity checks. The default is 10.

−−algorithm−texture−threshold=*value*
Sets a texture threshold value used to reject matches in low-texture image regions. The default is 10.

SEE ALSO

mbsystem(1), **mbgetphotocorrection**(1), **mbphotomosaic**(1), **mbimagelist**(1), **mbm_makeimagelist**(1), **mbgrid**(1)

BUGS

Maybe. Maybe not.